Household Electricity Demand Forecasting

A Comparative Study of Statistical and Deep Learning Time-Series Models

Dataset	UCI Individual Household Electric Power Consumption
Data Volume	2,075,259 minute-level readings (Dec 2006 - Nov 2010)
Forecast Target	Global Active Power (kW), resampled hourly
Forecast Horizon	24 hours ahead
Models Compared	Seasonal Naive, ARIMA, SARIMA, LSTM, GRU
Tools	Python, pandas, statsmodels, scikit-learn, TensorFlow/Keras

This report documents an end-to-end time-series forecasting pipeline built on real-world household electricity consumption data, including data cleaning, exploratory analysis, statistical diagnostic testing, and a comparative evaluation of five forecasting approaches ranging from a simple seasonal baseline to deep recurrent neural networks.

1. Project Overview

Short-term electricity demand forecasting is a core problem in energy analytics, used for grid load balancing, smart-home energy management, and demand-response systems. This project builds a complete forecasting pipeline on the UCI Individual Household Electric Power Consumption dataset, which records minute-level electrical measurements for a single household in Sceaux, France, from December 2006 to November 2010.

The objective was to forecast the household's **Global Active Power** (kW) 24 hours ahead, using an hourly-aggregated version of the series, and to rigorously compare a classical statistical baseline against deep learning sequence models using both accuracy metrics and residual diagnostics — not just point-forecast error.

2. Dataset and Preprocessing

The raw dataset contains 2,075,259 minute-level readings across 9 columns (Date, Time, and seven electrical measurements). Approximately 25,979 rows (~1.25%) per measurement column were missing, mostly from short logging/outage gaps.

Preprocessing steps applied:

- Combined Date and Time columns into a proper datetime index and sorted chronologically.
- Converted all measurement columns to numeric, coercing invalid entries to NaN.
- Resampled Global_active_power to hourly mean, producing 34,589 hourly observations.
- Filled 421 remaining hourly gaps using time-based interpolation, then forward/backward fill for edge cases.
- Explicitly set the hourly frequency (freq='h') to keep statsmodels forecasting aligned to the correct timestamps.
- Split chronologically (no shuffling, to respect time order): 70% train (24,212 hrs, Dec 2006-Sep 2009), 15% validation (5,188 hrs), 15% test (5,189 hrs, Apr-Nov 2010).

Summary statistics of the hourly target series (kW):

Count	Mean	Std Dev	Min	25%	Median	75%	Max
34,589	1.09	0.90	0.12	0.34	0.81	1.58	6.56

The distribution is right-skewed (mean 1.09 kW well above the median 0.81 kW), consistent with a household that spends most hours at low baseline draw with occasional high-usage spikes from appliances, cooking, or heating.

3. Exploratory Data Analysis

The full four-year hourly series (below) shows strong recurring seasonal structure and a gradual downward drift in typical consumption over the observation window.

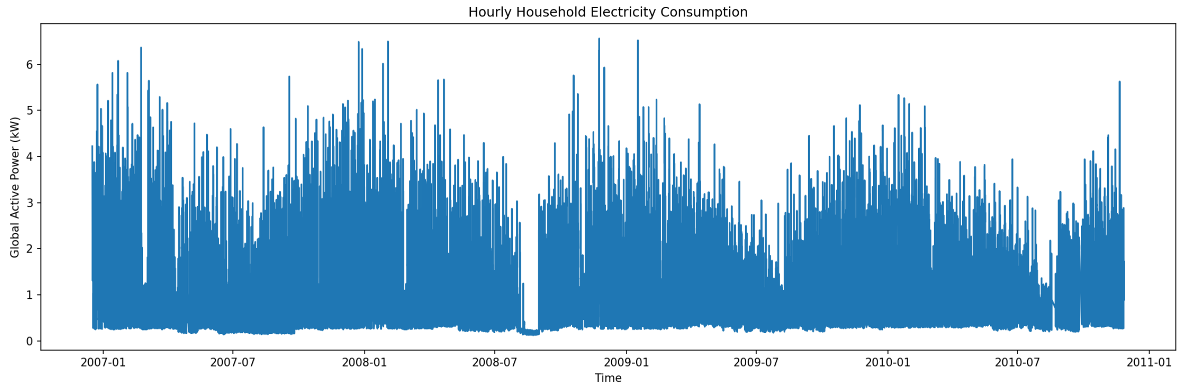

Figure 1: Full hourly household electricity consumption, Dec 2006 - Nov 2010.

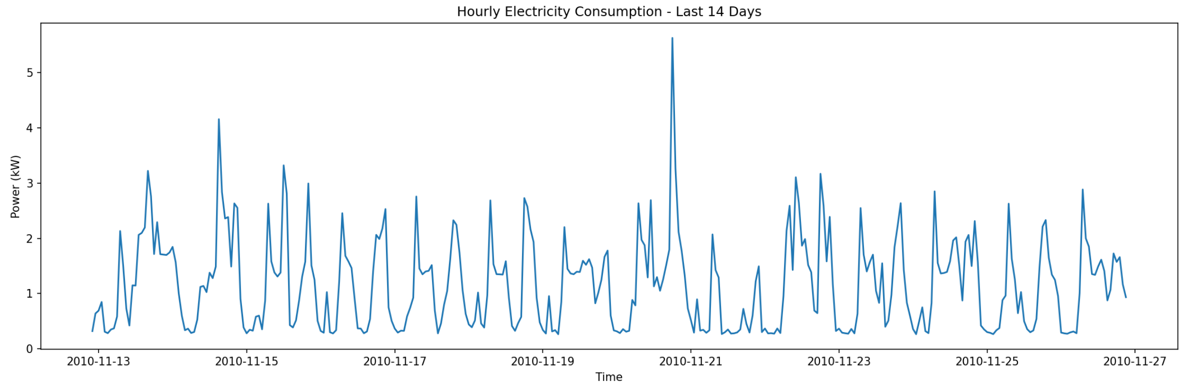

Figure 2: Last 14 days of the series, showing clear daily (24-hour) cyclical usage patterns.

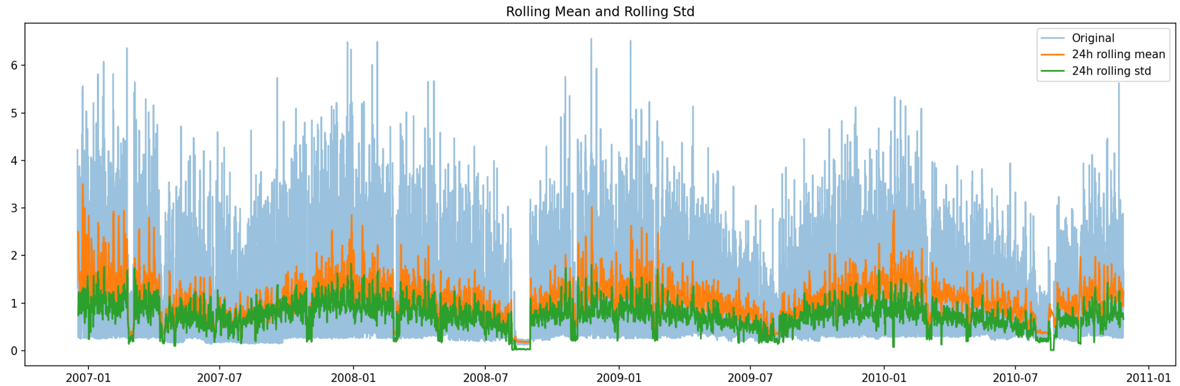

Figure 3: 24-hour rolling mean and standard deviation overlaid on the raw series, confirming the seasonal amplitude is roughly stable over time (no strong heteroscedastic drift).

3.1 STL Decomposition

The series (last 180 days of the dataset) was decomposed into trend, seasonal, and residual components using STL (Seasonal-Trend decomposition using LOESS).

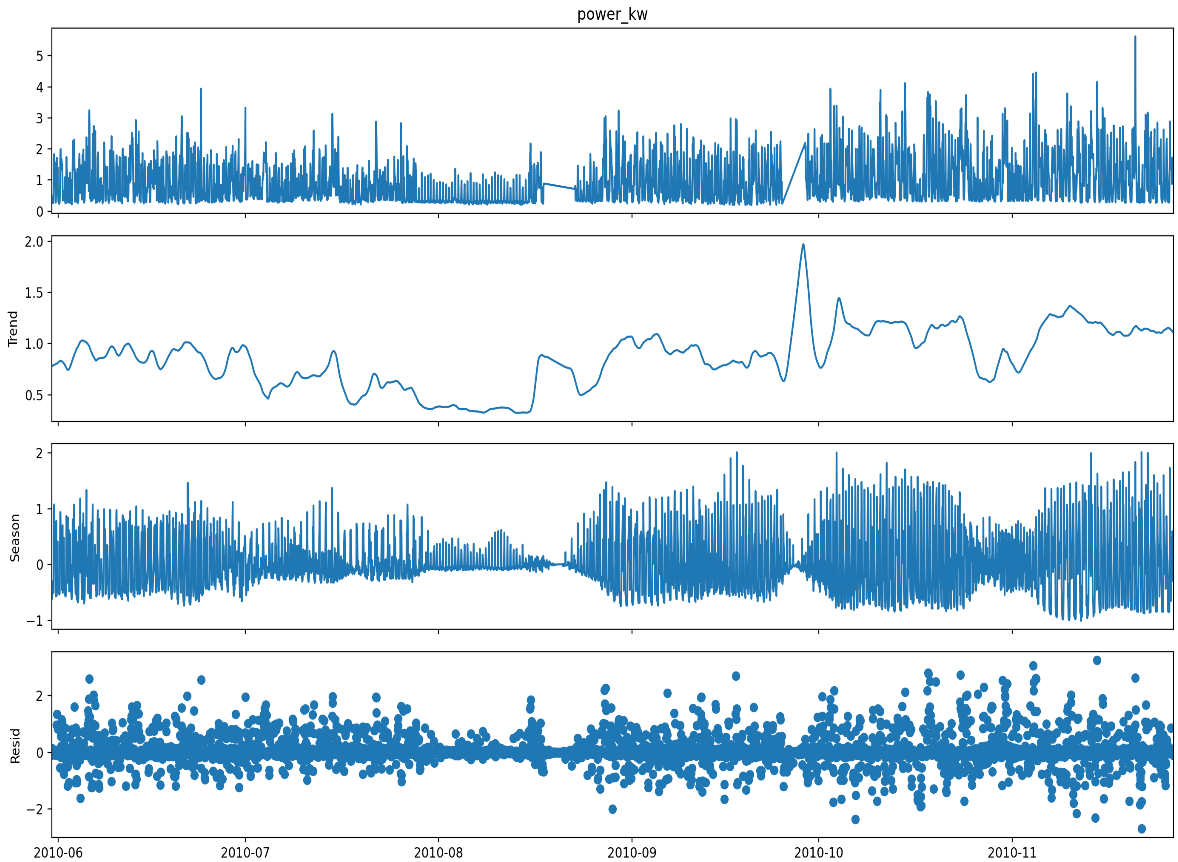

Figure 3B: STL decomposition of the last 180 days of hourly power consumption.

Two things stand out. First, the trend component shows a sharp, temporary spike around late September 2010, most likely a genuine short-term shift in household behavior (e.g. a period of unusually high occupancy or appliance use) rather than a data artifact, since it decays back to the prior level within a couple of weeks. Second, the seasonal component's amplitude clearly expands and contracts over time (larger daily swings around September-November, smaller through July-August) rather than staying constant — this amplitude modulation is consistent with the ARCH-LM test result in Section 7, which confirmed statistically significant volatility clustering rather than constant variance.

4. Statistical Diagnostics: Stationarity and Trend

Before model fitting, the training series was tested for stationarity and trend using three complementary tests, since no single test is fully conclusive on its own.

Test	Statistic	p-value	Conclusion
ADF (level)	-11.21	2.1e-20	Stationary evidence
KPSS (level)	2.11	0.01	Non-stationary evidence
Mann-Kendall (daily)	-4.79	1.6e-6	Significant decreasing trend (Sen's slope -0.000225/day)
ADF (1st difference)	-36.99	<0.001	Stationary
KPSS (1st difference)	0.032	>0.10	Stationary
ADF (seasonal diff, 24)	-34.34	<0.001	Stationary
KPSS (seasonal diff, 24)	0.012	>0.10	Stationary

Interpretation: ADF rejects the unit-root null while KPSS rejects the stationarity null — this is not a contradiction. It is the classic signature of a **trend-stationary** series: stationary in structure but with a small, statistically significant deterministic trend, which Mann-Kendall confirms directly (decreasing consumption over the 2006-2009 training window). After first-differencing, seasonal differencing (lag 24), and the combination of both, all series pass both ADF and KPSS as stationary — justifying the use of d=1 (and D=1 for SARIMA's seasonal component) in the subsequent ARIMA/SARIMA models.

4.1 ACF and PACF

Autocorrelation (ACF) and partial autocorrelation (PACF) were examined on a recent 60-day sample to guide model order selection.

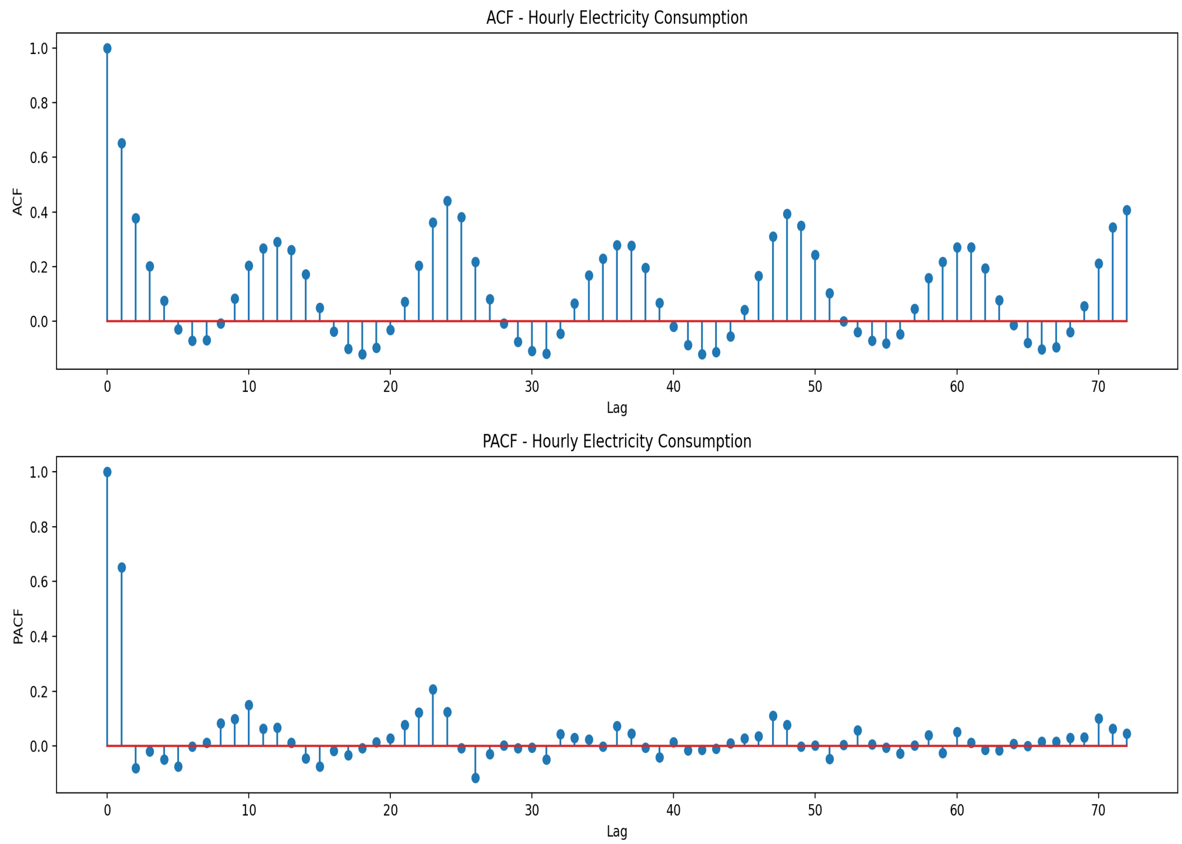

Figure 3C: ACF and PACF of hourly power consumption, up to 72-hour lags.

The ACF shows a slowly decaying pattern with clear repeating peaks at lags 24, 48, and 72 — direct confirmation of strong daily seasonality that must be handled explicitly (as SARIMA does) rather than left to a non-seasonal model. The PACF cuts off sharply after lag 1-2, with a secondary, smaller spike around lag 23-24. This pattern supports a low-order AR component for the non-seasonal part of the model (consistent with the AR(2) term used in ARIMA/SARIMA) plus a seasonal AR/MA term at lag 24, which is exactly the seasonal_order=(1,1,1,24) specification used for SARIMA in this project.

5. Forecasting Models

Seasonal Naive

Forecasts each hour using the value from the same hour exactly 24 hours earlier. Serves as the minimum-effort baseline any model must beat to justify its complexity.

ARIMA(2,1,2)

Fit on the most recent 10,000 training hours. Captures short-term autocorrelation and the first-order trend, but has no explicit mechanism for daily seasonality.

SARIMA(1,1,1)(1,1,1,24)

Extends ARIMA with an explicit seasonal component at lag 24 to directly model daily cyclicality, at the cost of more parameters and a more complex likelihood surface.

LSTM

A 2-layer LSTM (64 -> 32 units) with dropout, taking the previous 168 hours (7 days) as input to directly output all 24 forecast hours. 31,160 trainable parameters.

GRU

Architecturally parallel to the LSTM (64 -> 32 units, same lookback/horizon), using gated recurrent units for a lighter-weight alternative. 24,120 trainable parameters.

All models were evaluated using four standard forecasting metrics: MAE and RMSE (absolute error scale, kW), MAPE (percentage error, sensitive to near-zero actual values), and MASE (error scaled against a naive lag-1 baseline on the training set; MASE < 1 means the model beats naive lag forecasting).

6. Results

Model	MAE	RMSE	MAPE (%)	MASE
LSTM	0.435	0.592	62.94	0.658
GRU	0.480	0.625	75.25	0.726
ARIMA	0.622	0.725	116.52	0.940
Seasonal Naive	0.505	0.753	65.67	0.763
SARIMA	1.635	1.886	331.27	2.471

Table: Model performance on the held-out test period (Apr-Nov 2010). LSTM row highlighted as best performer by all four metrics.

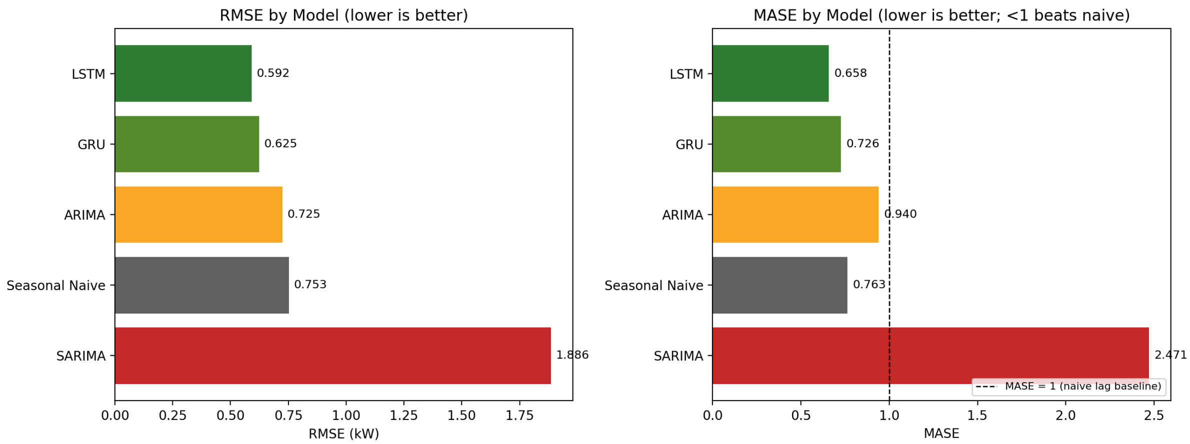

Figure 4: RMSE and MASE comparison across all five models. The dashed line in the MASE panel marks the naive-lag-1 baseline (MASE = 1).

Key findings:

- LSTM achieved the lowest error on every metric (RMSE 0.592 kW, MASE 0.658), a ~21% RMSE reduction versus the Seasonal Naive baseline and ~18% versus ARIMA.
- GRU performed close behind LSTM with roughly 30% fewer parameters, a reasonable efficiency trade-off.
- ARIMA modestly beat the naive baseline on MAE/RMSE but not on MAPE, reflecting its lack of an explicit seasonal term.
- SARIMA performed far worse than every other model, including the naive baseline (MASE 2.47). See the Limitations section below for why this happened and what it does — and doesn't — indicate.

6.1 Sample 24-Hour Forecast: LSTM vs GRU

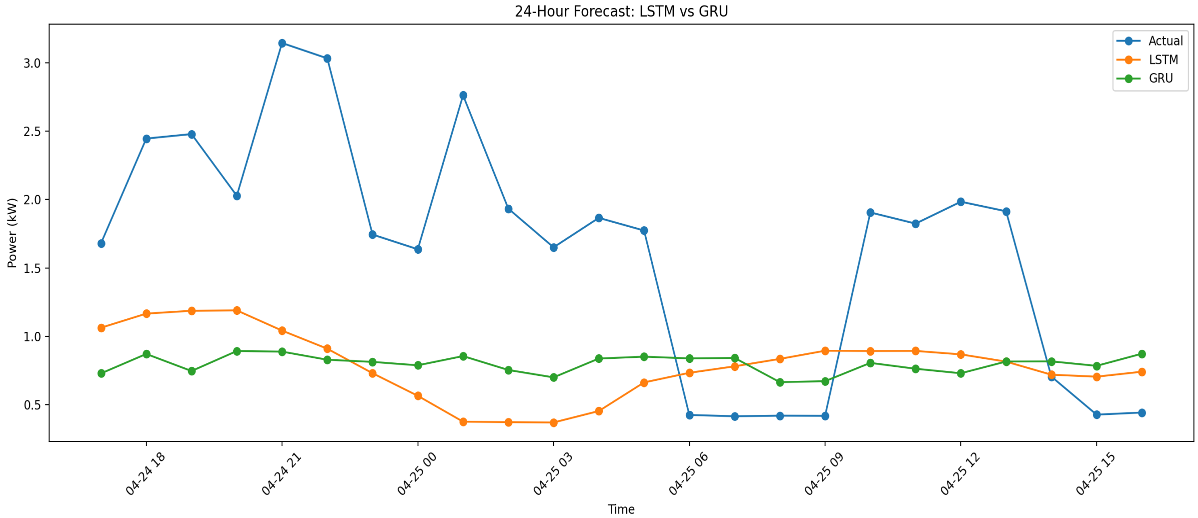

Figure 5: One 24-hour test window (24-25 April 2010), actual vs. LSTM vs. GRU predictions.

This individual window illustrates a broader pattern worth being transparent about: both models capture the household's low-baseline overnight hours reasonably well, but neither fully tracks the large demand spikes (actual reaching ~3.1 kW around 21:00 on 4/24 and ~2.8 kW around 01:00 on 4/25) — LSTM and GRU predictions stay closer to 0.7-1.2 kW through these peaks. This is a common and expected limitation of MSE-trained sequence models: minimizing squared error rewards smoothing toward the average behavior, so sudden appliance-driven spikes are systematically under-predicted. It does not invalidate the aggregate metrics reported above, but it does mean these models would need a different loss function (e.g. quantile loss) or spike-aware features to be reliable for peak-demand-sensitive applications.

6.2 Training Curves

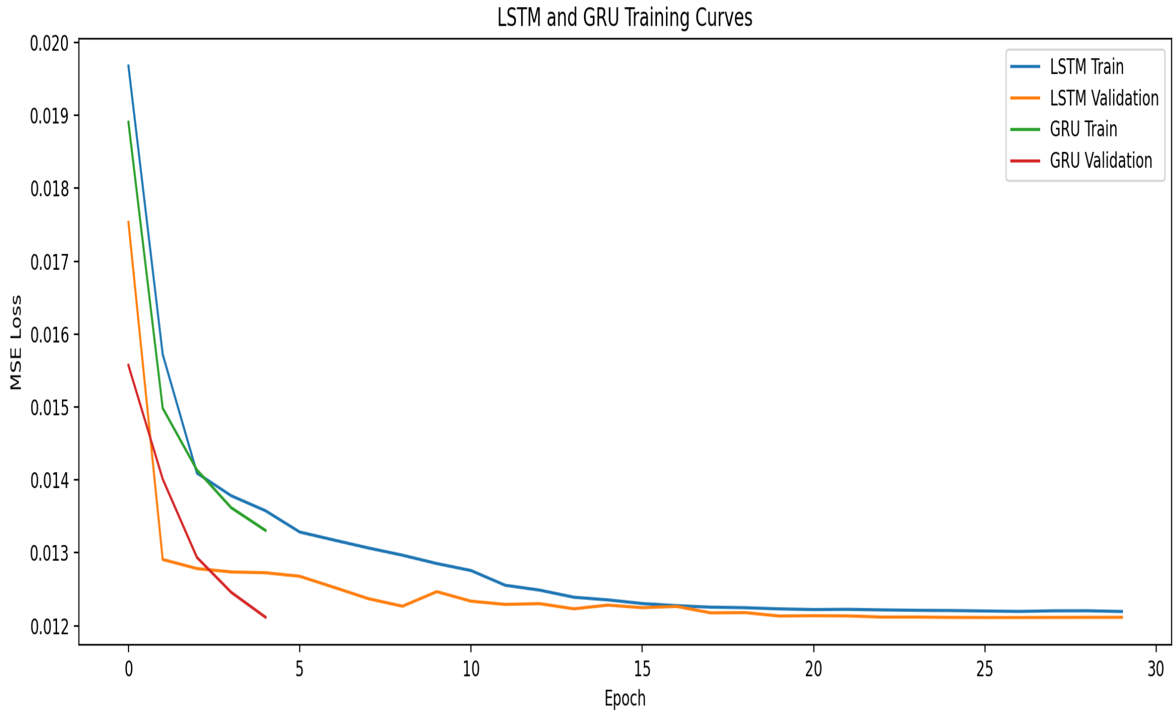

Figure 6: LSTM and GRU training and validation loss (MSE) by epoch.

Both models show smooth, well-behaved convergence with validation loss tracking training loss closely throughout — no evidence of overfitting. GRU's early-stopping callback triggered earlier (around epoch 5) than LSTM's (around epoch 28), reaching a similar final validation loss (~0.0121 MSE, scaled) in fewer epochs. This is consistent with GRU's simpler gating mechanism and roughly 30% fewer parameters (24,120 vs. 31,160) converging faster, even though LSTM's slightly longer training ultimately edged out a small accuracy advantage on the test set.

7. Residual Diagnostics

Point-forecast accuracy alone does not confirm a model is well-specified. Residuals (actual minus predicted) were tested for leftover autocorrelation (Ljung-Box) and volatility clustering (ARCH-LM).

Model	Residual Mean	Residual Std	Ljung-Box p	ARCH-LM p
ARIMA	-0.181	0.702	<0.001	<0.001
SARIMA	-1.525	1.110	<0.001	<0.001
LSTM	0.008	0.592	<0.001	<0.001
GRU	-0.042	0.623	<0.001	<0.001

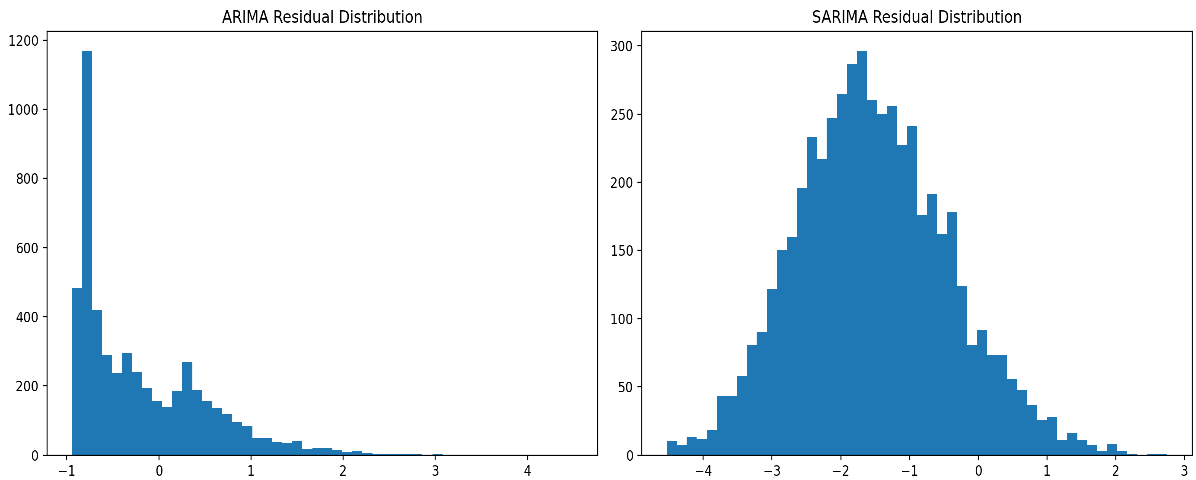

Figure 7: ARIMA and SARIMA residual distributions (test-period forecast errors).

The two histograms visually confirm what the residual means already suggested. ARIMA's residuals cluster in a sharp peak just below zero with a long right tail extending to +4 kW — the model mostly over-forecasts slightly (predicting a bit high) but occasionally misses badly on large demand spikes it cannot anticipate. SARIMA's distribution is shifted substantially further negative, centered around -1.5 to -2 kW, which is the direct visual signature of the systematic over-forecasting produced by its long-horizon unconditional drift discussed in Section 8.

Interpretation: Ljung-Box $p < 0.05$ for every model means residuals still carry statistically significant autocorrelation — none of the five models fully captures the series' temporal structure. ARCH-LM $p < 0.05$ for all models indicates volatility clustering (periods of large errors cluster together), consistent with the naturally spiky, appliance-driven nature of household power draw. LSTM's residual mean is closest to zero (0.008), indicating the least systematic bias, while SARIMA's large negative residual mean (-1.525) confirms it is not just noisier but is now systematically over-forecasting the series.

8. Limitations and Notes on Methodology

Evaluation protocol is not fully matched across models. ARIMA and SARIMA were fit once and asked to forecast the entire ~216-day test period unconditionally (no access to actual values during the forecast horizon). LSTM and GRU, by contrast, were evaluated on many overlapping 24-hour windows, each conditioned on the true 168 hours of history immediately preceding it. This gives the deep learning models a substantial informational advantage and is the primary reason SARIMA's error compounds so severely over the long unconditional horizon (its seasonal MA coefficient, near -0.96, amplifies small parameter imprecision over hundreds of forecasted days). A fair, publication-quality comparison requires a common rolling-origin (walk-forward) evaluation protocol for all five models, refitting or updating ARIMA/SARIMA on real observed values every 24 hours, the same way the neural models were effectively evaluated. This is the direct next step for the project.

MAPE is unreliable here. Because many actual hourly values sit close to zero kW (low-draw overnight hours), small absolute errors translate into very large percentage errors. MAPE values in this report (62-331%) should be read as a secondary, not primary, accuracy signal — MAE, RMSE, and MASE are more trustworthy for this dataset.

No prediction intervals were produced. Given confirmed volatility clustering (ARCH-LM), constant-variance confidence intervals would be misleading; a GARCH-type variance model or quantile-based deep learning approach would be needed for calibrated intervals.

9. Conclusion

This project delivered a complete, statistically grounded electricity-demand forecasting pipeline: from raw minute-level sensor data to a validated comparison of five models. The LSTM model produced the most accurate 24-hour-ahead forecasts by every metric tested, with the GRU a close, more lightweight second. The classical ARIMA model provided a useful, interpretable middle ground, while SARIMA's unconditional long-horizon instability highlights a genuinely important lesson in forecasting methodology: matching the evaluation protocol across models is as important as the choice of model itself.

10. Tools and Techniques Used

- Data handling: pandas, numpy
- Statistical testing: statsmodels (ADF, KPSS, Ljung-Box, ARCH-LM, STL), pymannkendall (Mann-Kendall, Sen's slope)
- Classical forecasting: statsmodels ARIMA / SARIMAX
- Deep learning: TensorFlow / Keras (LSTM, GRU)
- Evaluation: scikit-learn (MAE, RMSE), custom MAPE/MASE implementations
- Visualization: matplotlib, seaborn

Dataset source: UCI Machine Learning Repository - Individual Household Electric Power Consumption. Full code, figures, and saved model artifacts are available in the project repository.